5M-Building: A Large-Scale High-Resolution Building Dataset with CNN based Detection Analysis

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***Abstract*—Building detection in remote sensing images plays an important role in applications such as urban management and urban planning. Recently, convolutional neural network (CNN) based methods which benefits from the popularity of large-scale datasets have achieved good performance for object detection. To our best knowledge, there is no large-scale remote sensing image dataset specially build for building detection. Existing building datasets are in small size and lack of diversity, which hinder the development of building detection. In this paper, we present a large-scale high-resolution building dataset named 5M-Building after the number of samples in the dataset. The dataset consists of more than 10 thousand images all collected from GaoFen-2 with a spatial resolution of 0.8 meter. We also present a baseline for the dataset by evaluating three state of the art CNN based detectors. The experiments demonstrate that it is great challenge to accurately detect various buildings from remote sensing images. We hope the 5M-Building dataset will facilitate the research on building detection.**

Keywords—Building dataset, Building detection, High-resolution images, Smart city.

I. INTRODUCTION

In recent years, the application of high-resolution remote sensing images has achieved considerable progress and gained great success in research filed such as urban management and planning. High-resolution remote sensing images have the advantages of high spatial resolution and high temporal resolution, which facilitate applications for researches in land [1], housing, forestry [2], ocean, transportation [3] and military. In these research areas, target detection [4] and classification [5] are key steps for understanding and interpretation of remote sensing images, and continuously play an important role.

Nowadays, more and more remote sensing satellites equipped with high spatial resolution imaging sensors are launched, providing researchers amount of high quality images available for interested objects detection. Among these objects, buildings are of greatly important yet have attracted a few attention from community notwithstanding most of ground man-made objects are buildings. Over past few years, various large scale remote sensing datasets are annotated and released to community for research. These datasets are either focus on land cover classification or generic ground objects detection. Although there are building datasets such as UC Merced land-use Dataset [6], WHU-RS19 Dataset [7], RSSCN7 Dataset [8], SIRI-WHU Dataset [9] and NWPU-RESISC45 Dataset [10], these dataset are mainly annotated for building roof extraction i.e. only binary roof masks are provided and more importantly they are in small size and lack of diversity. It has been widely accepted that state of the art performance is achieved by deep learning based methods which require huge amount of labeled data for training. Thus, it is critical to build large scale building dataset. In addition to this, almost all existing datasets collect images from Google Earth. Although the spatial resolution of these images is relatively high, the quality of them is poor, which hardly produce the best results.

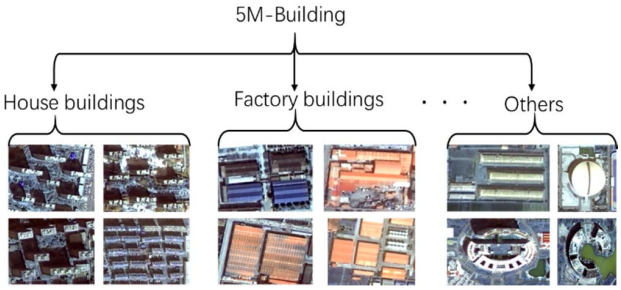

Fig. 1. Overview of 5M-Building.

These issues bring challenges to problem of detecting building from remote sensing images with deep learning techniques. To Address these issues, in this work a large-scale high-resolution building dataset is annotated. We also present a brief experimental analysis for building detection using the proposed dataset. The dataset consists of more than 5 million buildings so we call it 5M-Building dataset. Images in the 5M-Building dataset are acquired by GaoFen-2 high-resolution satellite of China. Fig.1 shows several samples of 5M-Building. More than 10,000 remote sensing images were collected and all buildings in images were manually annotated by people who have background of Geoscience and Remote Sensing. Buildings in this dataset have different functions and architectural styles and thus are in diversity scales and appearances as shown in Fig. 1. We build a baseline for building detection by conducting experiments on the proposed 5M-Building dataset with several state of the art deep learning based algorithms. The experiments demonstrate that detecting buildings from remote sensing images is very challenging, further researches are encouraged to promote real application of building detection.

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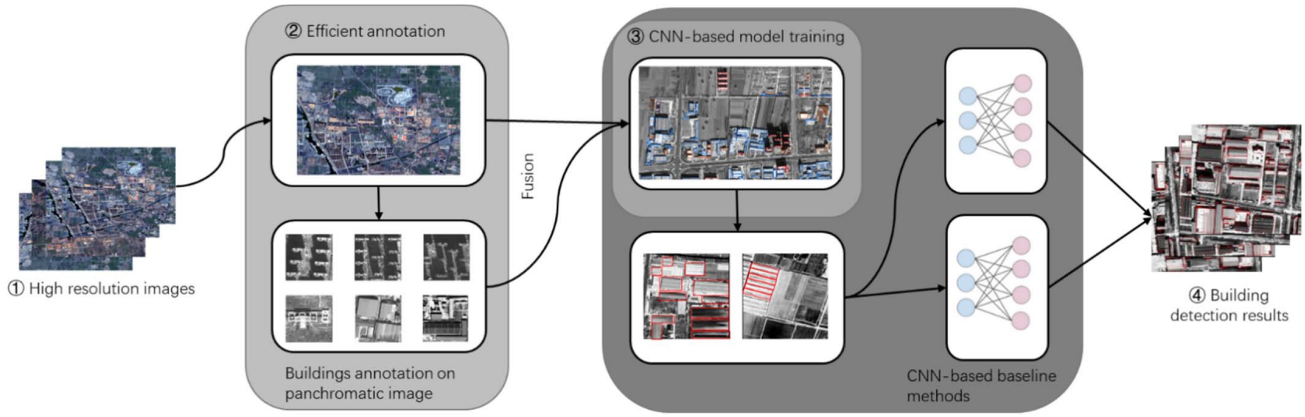


Fig. 2. The framework of our work. Firstly, we collect high resolustion images acquired by GaoFen-2 satellite. Sencondly, annotations are conducted on panchromatic images. Before training, pancnromatic and multispectral image are fused to generate high spatial resolution multispectral images for better detection. We employ CNN based mehtods to obtain building detection results.

The main contributions of this paper are: (1) a large scale high-resolution building dataset for building detection in remote sensing images is proposed. To our best knowledge, this is the largest building dataset. (2) We build a baseline with sate of the art CNN detectors for further research.

The rest of the paper is organized as follows: we review and summarize existing building datasets and building detection methods in Section 2. The proposed 5M-Building dataset is introduced in Section 3. Section 4 presents the experiments with analysis and conclusions are given in Section 5.

II. Related work

Detecting or recognizing land objects is a key step for interpretation of remote sensing images, thus has been receiving increasing attention from remote sensing community. A variety of datasets involving this topic have been released. In this work, we are interest in buildings and mainly focus on detecting or locating them from images. We review datasets that contain buildings and summarize them in Table I. It should be noted that many of them consist of object categories more than buildings. In this paper, we only list statistical information of buildings.

Many remote sensing object detection methods have also been proposed by researchers. In recent years, deep learning based generic object detection have achieved great success, inspiring researchers in remote sensing community to design object detection methods based on deep learning techniques. For example, Ref [11] proposed an end-to-end target heat-map network (THNet) for target detection in remote sensing images. The THNet is a fully convolutional layer, thus it has no limitation in the size of the input images and the output of which is a heated layer of the target. A new detection paradigm called 'random access memories(RAM)' was proposed by Zou et al. [12]. Their method achieved significant improvements over CNN based detectors and was superior to several other methods. The RSI saliency target detection algorithm based on sparse reconstruction was proposed by Jing et al. [13]. Their method could accurately distinguish the foreground and background by combining low-level and high-level visual information, which is an unsupervised method that can be used to detect common targets and has strong generalization capabilities.

Although researchers have built many datasets for remote sensing target detection and classification and made great progress in target detection and classification algorithms, challenges such as insufficient training samples, lack of diversity of samples, low quality images have slow down progress in this research field. To address these problems, this paper first constructs a building dataset named 5M-Building. Compared with the existing datasets, the proposed dataset has the advantages of high-resolution, large-scale and diversity of building samples, which could significantly facilitate the research of target detection in remote sensing images. Then, we present a baseline evaluation for the dataset using several widely used CNN based detection methods. Fig.2 show the framework of our work, in which we illustrated how we construct the proposed dataset.

TABLE I. Remote sensing image datasets that contain buildings

Dataset	Information				
	Sources	***Image size***	***Number of building samples***	***Spatial resolution(m)***	***Year***
UC Merced land-use[6]	USGS	256*256	500	0.3	2010
WHU-RS19[7]	Google earth	600*600	150	0.5	2012
RSSCN7[8]	Google earth	400*400	800	--	2015
SIRI-WHU[9]	Google earth	200*200	600	2	2016
NWPU-RESISC45 [10]	Google earth	256*256	5600	~30 to 0.2	2016
5M-Building	GaoFen	512*512	10000	0.8	2019

III. 5M-BUILDING

5M-Building dataset is constructed via manually annotation completed by people who have strong background in geoscience and remote sensing, in which images are with high spatial resolution and acquired by GaoFen-2 satellite. Details of the 5M-Building are described below.

A. Annotation

Buildings are with different functions. In our work, we divide buildings into house buildings, factory buildings and other buildings, etc. The images are acquired by GaoFen-2 satellite covering Shandong province of China and with a spatial resolution of 0.8 meter for panchromatic image and 3.2 meter for multispectral image. The size of images ranges from 2000×2000 and 5000×5000 for annotation convenience.

During annotation, we ask labelling works to annotate the geometric outlines of buildings which consistent with the contours of them. The annotations are conducted on the panchromatic images. Because the annotations are stored in shape vector format, they can directly applied on the corresponding multispectral images. And we can easily generate bounding boxes from the annotations.

B. Samples in 5M-Building

Buildings in 5M-Building dataset consists of three distinct categories, i.e., residential buildings, factory buildings and other buildings. Because the images are too large to fed into CNN detectors, we crop them into 512×512 sub-images. This will generate more than 10,000 samples. Samples of residential buildings, factory buildings and other buildings are illustrated in Fig. 3.

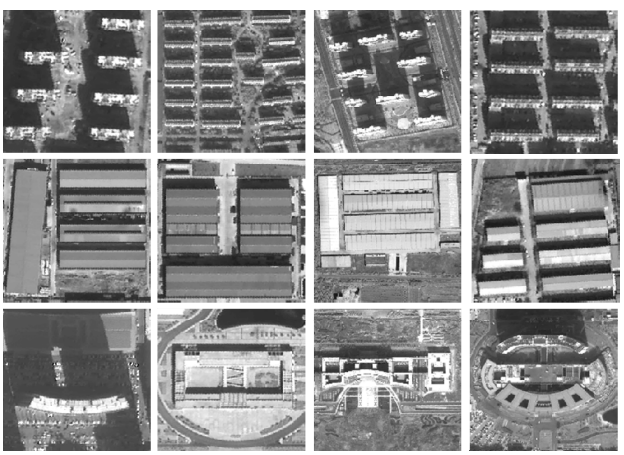


Fig. 3. Sample images of buildings. Residential buildings are shown in the top row, factory buildings are presented in midle row and others are illustrated in the bottom row.

These images are acquired under various conditions of different weathers and seasons. Buildings in the dataset are also covers characteristics of scale changes, intra-class diversities and inter-class similarities. Compared with many existing remote sensing datasets that focus on land covers and scene classification, we intend to present a large-scale dataset built only for building detection. To the best of our knowledge, the 5M-Building dataset proposed in this paper is the largest high-resolution remote sensing dataset for building detection.

IV. EXPERIMENTAL RESULT AND ANALYSIS

In this section, we present a baseline for the 5M-Building dataset by evaluating three commonly used deep neural networks i.e., Faster R-CNN [14], Mask R-CNN [15] and YOLO v3 [16].

A. Faster R-CNN

In our experiments, we employ Faster R-CNN equipped with RPN for evaluation, the backbone of which is ResNet-50. The hyper parameters are set as follows: anchor_stride=(4, 8, 16, 32, 64), anchor_sizes=(8, 16, 32, 64, 128), score_thresh=0.5, base_lr=0.01, weight_decay=0.0001, min_size_train=(512, 576, 640, 704, 800), min_size_test=800.

The loss function for Faster R-CNN is:

$$\mathrm{L}(\{P_i\},\{t_i\}) = \frac{1}{N_{cls}}\sum_i L_{cls}\,(p_i, p_i^*) + \lambda\frac{1}{N_{reg}}\sum_i p_i^*\,L_{reg}(t_i, t_i^*) \quad (1)$$

where P_i is the probability of an anchor predicted as the target, p_i^* is GT label, $L_{cls}(p_i, p_i^*)$ is the logarithmic loss of the target and non-target, t_i represents the deviation predicted by the anchor, t_i^* represents the actual deviation of anchor relative to GT, and N_{reg} is number of objects.

Table II shows the experimental result. The loss curve of Faster R-CNN with respect to training iteration is shown in Fig. 4.

TABLE II. EXPERIMENTAL RESULTS OF FASTER R-CNN

IoU	*size*	*Average Precision (AP)*	*Average Recall (AR)*
0.50:0.95	all	0.260	0.388
0.50	all	0.508	--
0.75	all	0.238	--
0.50:0.95	small	0.183	0.300
0.50:0.95	medium	0.403	0.556
0.50:0.95	large	0.471	0.632

B. Mask R-CNN

The parameters of Mask R-CNN are set as follows: conv_body=r-50-fpn, anchor_stride=(4, 8, 16, 32, 64), anchor_sizes=(8, 16, 32, 64, 128), score_thresh=0.5, base_lr=0.005, weight_decay=0.0001, min_size_train=(512, 576, 640, 704, 800), min_size_test=800.

The loss function L for Mask R-CNN is:

$$\mathrm{L} = L_{cls} + L_{box} + L_{mask} \quad (2)$$

In Equation 2, L_{mask} denotes the classification result of each pixel. It is defined as an average binary cross-entropy loss. The experimental results of Mask R-CNN are shown in

Table III, and the loss function curve is demonstrated in Fig. 5.

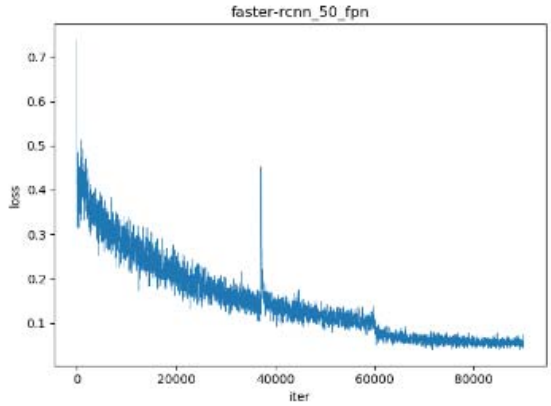


Fig. 4. The loss function curve of Faster R-CNN.

TABLE III. EXPERIMENTAL RESULTS OF MASK R-CNN

IoU	*area*	*Average Precision (AP)*	*Average Recall (AR)*
0.50:0.95	all	0.232	0.368
0.50	all	0.473	--
0.75	all	0.208	--
0.50:0.95	small	0.142	0.286
0.50:0.95	medium	0.390	0.526
0.50:0.95	large	0.476	0.554

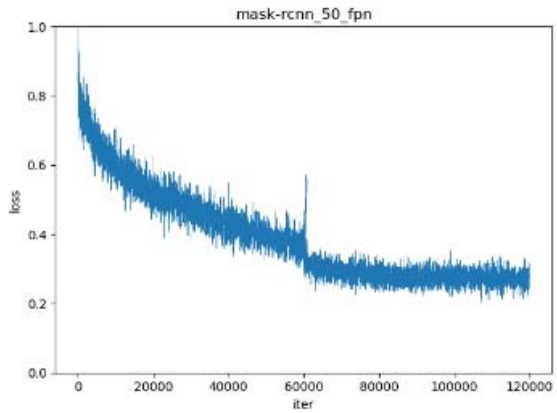


Fig. 5. The loss function curve of Mask R-CNN.

C. YOLO v3

Parameters of YOLO v3 are set as follows: width=608, height=608, momentum=0.9, decay=0.0005, angle=0.2, learning_rate=0.001, burn_in=1000, max_batches=90000, steps=60000, 80000, anchors=10, 8, 14, 17, 23, 51, 26, 9, 32, 20, 59, 17, 64, 32, 105, 78, 112, 24. The final experimental results of YOLO v3 are shown in Table 4 and the loss function curve is shown in Fig. 6.

Experiments results show the validity of 5M-Building dataset.

TABLE IV. EXPERIMENTAL RESULTS OF YOLO V3

IoU	*area*	*Average Precision (AP)*	*Average Recall (AR)*
0.50:0.95	all	0.268	0.340
0.50	all	0.479	--
0.75	all	0.275	--
0.50:0.95	small	0.207	0.276
0.50:0.95	medium	0.383	0.470
0.50:0.95	large	0.319	0.372

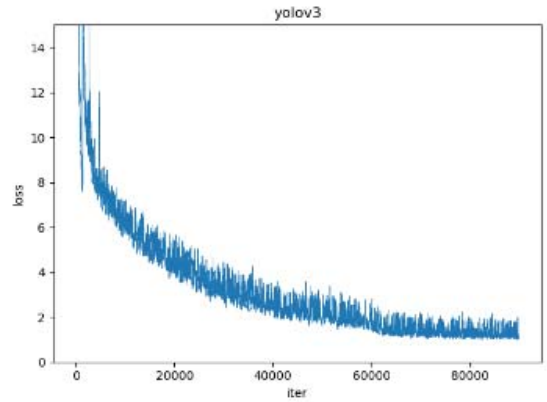


Fig. 6. The loss function curve of YOLO v3.

D. Building detection results

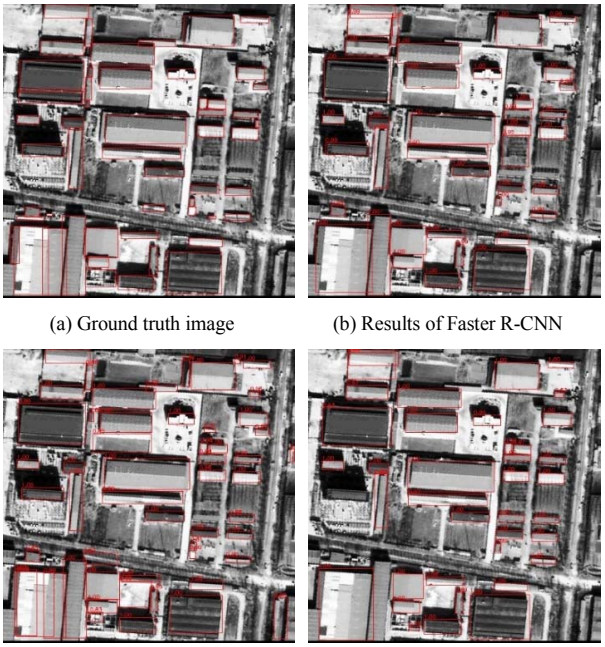
The trained models are utilized for building detection of remote sensing images. The detection results are illustrated in Fig. 7. The ground truth image with bounding boxes is showed in Fig. 7(a), red bounding boxes indicate buildings in ground truth image. Fig. 7(b) to Fig. 7(d) show the building detection results of Faster R-CNN, Mask R-CNN and YOLO v3, respectively. As demonstrated in Fig. 7, the commonly used target detection methods present satisfactory detection results for residential buildings and factory buildings.

Residential buildings and factory buildings have salient appearance features, and these buildings usually exist in or near cities. The models could accurately detect these buildings in 5M-Building. And as shown in Fig. 7, Mask R-CNN achieves the best detection results. 5M-Building proposed in this paper only contains basic buildings which will be expansion to contain more samples in future. Dense building area is not a category of 5M-Building. However, Faster R-CNN and Mask R-CNN also achieve better building detection result for buildings in dense area. Fig. 8(b) and Fig. 8(c) show buildings detection results via Faster R-CNN and Mask R-CNN.

V. CONCLUSION

In this paper, a large-scale building dataset named 5M-Building is proposed. Three CNN based methods, i.e., Faster R-CNN, Mask R-CNN and YOLO v3 were utilized for

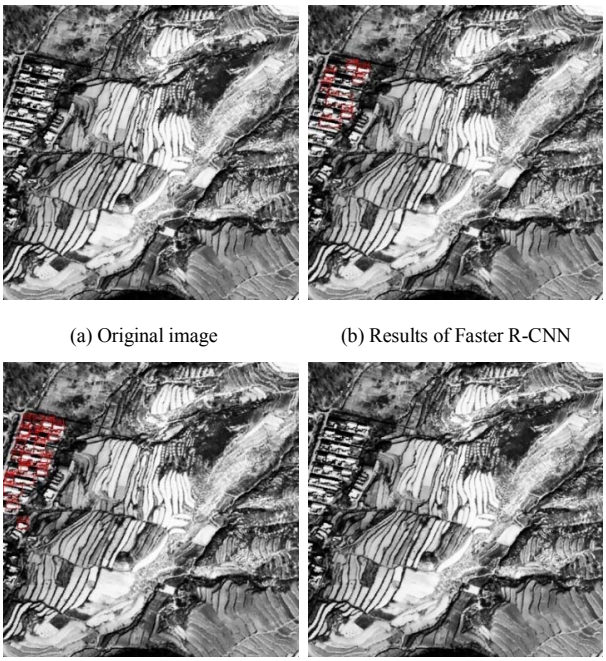
evaluation of 5M-Building. The proposed dataset contains more than five million building individuals. Building samples in dataset are abundant and diverse, with features of large-scale, multi-scale, intra-class diversity and inter-class similarity in building.



(a) Ground truth image (b) Results of Faster R-CNN

(c) Results of Mask R-CNN (d) Results of Yolo v3

Fig. 7. The building detection results of Faster R-CNN, Mask R-CNN and YOLO v3.



(a) Original image (b) Results of Faster R-CNN

(c) Results of Mask R-CNN (d) Results of Yolo v 3

Fig. 8. The building detection results of dense buildings.

The proposed dataset and building detection methods derived from 5M-Building will play an important role in applications of smart city, such as urban management and planning. In the future work, we will further expand this dataset and propose effective building detection methods.

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REFERENCES

[1] Z. Gong, P. Zhong, W. Hu, and Y. Hua, "Joint Learning of the Center Points and Deep Metrics for Land-Use Classification in Remote Sensing," Remote Sens., vol. 11, p. 76, 2019.

[2] B. Bai, Y. Tan, D. Guo, and B. Xu, "Dynamic Monitoring of Forest Land in Fuling District Based on Multi-Source Time Series Remote Sensing Images," ISPRS Int. J. of Geo-Inf., vol. 8, p. 36, 2019.

[3] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual u-net," IEEE Geosci. Remote Sens. Lett., vol. 15, pp. 749-753, 2018.

[4] Q. Li, L. Mou, Q. Liu, et al, "HSF-Net: Multiscale deep feature embedding for ship detection in optical remote sensing imagery," IEEE Trans. Geosci. Remote Sens., 2018, vol. 56, pp. 7147-7161.

[5] W. Li, H. Liu, Y. Wang, Z. Li, Y. Jia, and G. Gui, "Deep Learning-Based Classification Methods for Remote Sensing Images in Urban Built-Up Areas," IEEE Access, vol. 7, pp. 36274-36284, 2019.

[6] Q. Zhu, Y. Zhong, B. Zhao, G.-S. Xia, and L. Zhang, "Bag-of-visual-words scene classifier with local and global features for high spatial resolution remote sensing imagery," IEEE Geosci. Remote Sens. Lett., vol. 13, pp. 747-751, 2016.

[7] F. P. Luus, B. P. Salmon, F. Van den Bergh, and B. T. J. Maharaj, "Multiview deep learning for land-use classification," IEEE Geoscience and Remote Sensing Letters, vol. 12, pp. 2448-2452, 2015.

[8] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," IEEE Geosci. Remote Sens. Lett., vol. 12, pp. 2321-2325, 2015.

[9] B. Zhao, Y. Zhong, G.-S. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," IEEE Trans. Geosci. Remote Sens., vol. 54, pp. 2108-2123, 2016.

[10] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," Proc. of the IEEE, vol. 105, pp. 1865-1883, 2017

[11] H. Chen, L. Zhang, J. Ma, and J. Zhang, "Target heat-map network: An end-to-end deep network for target detection in remote sensing images," Neurocomputing, vol. 331, pp. 375-387, 2019.

[12] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," IEEE Trans. on Image Process., vol. 27, pp. 1100-1111, 2017.

[13] M. Jing, D. Zhao, Z. Jiang, and L. Li, "Proposal based saliency model for generic target detection in remote sensing image," in 2017 IEEE RCAR, 2017, pp. 309-314.

[14] W. Zhang, "Characters Detection on Namecard with faster RCNN," arXiv preprint arXiv:1807.10417, 2018.

[15] A. O. Vuola, S. U. Akram, and J. Kannala, "Mask-RCNN and U-net Ensembled for Nuclei Segmentation," arXiv preprint arXiv:1901.10170, 2019.

[16] J. Redmon and A. Farhadi, "Yolov3: An incremental improvement," arXiv preprint arXiv:1804.02767, 2018.